Paleomagnetism of the Pine Point Zn–Pb deposits[1]

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Received October 5, 1992
Revision accepted February 15, 1993

The Pine Point Zn–Pb deposits are Mississippi Valley-type (MVT) deposits in Middle Devonian dolostone of the Pine Point Formation. The median natural remanent magnetization intensity from 509 specimens from 40 sites is 3.6×10^{-5} A/m. This intensity is an order of magnitude lower than most MVT deposits and is close to the sensitivity of the cryogenic magnetometer, so only the stronger specimens could be analysed paleomagnetically. Isothermal remanent magnetization testing of ores, host rocks, and monomineralic specimens shows that single domain to pseudosingle domain magnetite in sphalerite is the main remanence carrier; magnetite in galena is the next most significant carrier. In contrast, barren dolostone, limestone, and calcite crystals have a much weaker remanence. Using less stringent criteria than is conventional, 17 acceptable site mean remanence directions were determined. Remagnetization circle data indicate that many of the poorly defined sites retain a reversed but nonisolated characteristic remanence. Of the 17 means, 8 form a coherent cluster, the A component, with a mean direction of $D = 339°$, $I = 78°$ ($\alpha_{95} = 7°$, $k = 62$), giving a pole position of 79°N, 197°E ($\alpha_{95} = 13°$). The A pole indicates with >95% confidence that the MVT ores at Pine Point are Late Cretaceous to Eocene in age, with a mid-Late Cretaceous to Paleocene age (84–58 Ma) being most likely. The dolostone host rocks were remagnetized at the same time. It is concluded that the A component, carried mainly by sphalerite and galena, was formed during the Laramide orogeny (97–58 Ma) when tectonic uplift forced Zn–Pb-rich basinal brines out of the Western Canada sedimentary basin into the Presqu'ile barrier reef on the adjacent foreland shelf.

Les gisements de Zn–Pb de Pine Point sont du type de la vallée du Mississippi (TVM), incorporés dans les dolomies d'âge dévonien moyen de la Formation de Pine Point. L'intensité moyenne de l'aimantation rémanente naturelle de 509 spécimens, provenant de 40 sites, est $3{,}6 \times 10^{-5}$ A/m. Cette intensité correspond à un ordre de magnitude inférieure à celles obtenues pour la majorité des gisements de TVM, et peu éloignée de la limite de sensibilité du magnétomètre cryogénique, par conséquent les analyses paléomagnétiques de ces gisements ne sont possibles que pour les spécimens à intensité la plus élevée. Les essais demesure de l'aimantation rémanente isothermique sur les minerais, les roches encaissantes de la minéralisation, et les spécimens monominéralogiques révèlent que la rémanence principale est portée par les domaines simples ou pseudo-simples de la magnétite incluse dans la sphalérite; la magnétite dans la galène occupe le second rang en importance comme porteur de la rémanence. D'autre part, la dolomie stérile, la calcaire et les cristaux de calcite possèdent une rémanence très faible. En appliquant des critères moins rigoureux que ceux utilisés habituellement, on a déterminé des moyennes acceptables de 17 sites de directions de rémanence. Les données du cercle de réaimantation indiquent que plusieurs des sites pauvrement définis retiennent une rémanence inversée, mais non isolée typiquement. Parmi ces 17 moyennes, 8 forment un groupe cohérent, la composante A, avec une direction moyenne de $D = 339°$, $I = 78°$ ($\alpha_{95} = 7°$, $k = 62$), qui donne une position de pôle de 79°N, 197°E ($\alpha_{95} = 13°$). Le pôle A indique avec >95% de confiance que les minerais de TVM à Pine Point sont d'âge crétacé tardif à éocène, et plus probablement avec un âge entre le milieu du Crétacé tardif et le Paléocène (84–58 Ma). À cette même époque, les roches hôtes de dolomie furent réaimantées. La conclusion est que la composant A, portée principalement par la sphalérite et la galène, fut acquise durant l'orogenèse du Laramide (97–58 Ma), lorsque le soulèvement tectonique a forcé l'évacuation des saumures riches en Zn–Pb des dépressions hors du bassin sédimentaire de l'Ouest du Canada vers le récif barrière de Presqu'île sur la plate-forme adjacente de l'avant-pays.
[Traduit par la rédaction]

Can. J. Earth Sci. **30**, 1028–1036 (1993)

Introduction

This paper reports results from an extensive paleomagnetic study aimed at dating the Mississippi Valley-type (MVT) zinc–lead deposits at Pine Point in the Northwest Territories, Canada (Fig. 1). The age of these ores is essentially unconstrained by either geologic or radiometric evidence. The main problem faced was that the remanence intensities of these ores and host rocks are about an order of magnitude weaker than typical for MVT ore districts. This was first reported by Beales et al. (1974), and it makes it difficult to define the stable remanent magnetization direction. Successful determination of the age provides a critical test of several models for the genesis of MVT deposits.

Geology

The Pine Point ore deposits are hosted by Givetian dolostone (377 ± 3 Ma) of the Presqu'ile barrier complex in the Middle Devonian Pine Point Formation (Rhodes et al. 1984). About half of the 90 or so known mineralized zones were mined within an area of about 200 km^2, between 1964 and 1988, yielding 64.3×10^6 t of ore grading 7.0% Zn and 3.4% Pb, confirming it as one of the world's major MVT districts. The Pine Point Formation has dips of <1° throughout the study area. The mineralization is comprised of sphalerite, galena,

[1]Geological Survey of Canada Contribution 17593.

Printed in Canada / Imprimé au Canada

marcasite, pyrite, and dolomite. It occurs as cement in collapse breccias or replacements associated with anastomosing paleokarst channelways distributed in two linear networks trending about N65°E, known as the Main and North trends (Fig. 1). Both trends contain interconnected horizontal tabular and vertical chimney-like mineralized zones.

The age of these ores has proved the enigma. Karstification began prior to deposition of the overlying Middle Devonian Watt Mountain Formation, and it continued possibly as late as the start of Pleistocene time (Rhodes et al. 1984). Cumming et al. (1990) report Pb isotopic ratios on 32 galena samples. In order of preference the authors obtained an age of about 290 Ma using a single-stage lead-evolution model, about 60 Ma using a three-stage model, and about 375 Ma using a mixing model. The correct choice of model is uncertain because the galenas are remarkably homogeneous.

Thermal evidence combined with burial-depth information provide an indirect measure of the age of the Pine Point deposits. Fluid-inclusion data indicate that the ore minerals were precipitated at temperatures ranging from 50 to 100°C, with an average of about 85°C (Roedder 1968; Kyle 1981). Reflectance studies on bitumen from Pine Point and nearby liptinite-rich coals indicate maximum postdepositional temperatures of about 60–100°C for the host rocks (Goodarzi and MacQueen 1990). Qing and Mountjoy (1990, 1992) used petrographic and oxygen-isotope and other geochemical analyses to identify four types of dolomite at Pine Point. They found that the late Presqu'ile saddle dolomite, which is most intimately associated with the Zn–Pb ore, requires a formation temperature of at least 90°C. These temperatures appear to have been reached only during the Laramide orogeny. Fission track studies on apatite by Arne (1991) show that Pine Point was not heated to temperatures over 80°C prior to Late Cretaceous burial. Regional heat flow studies by Majorowicz et al. (1989) show that Pine Point has an average geothermal gradient for the Western Canada sedimentary basin, so the only time when hydrothermal fluid temperatures could be sufficiently high to produce the Pine Point MVT ores would be during the Late Cretaceous or early Tertiary.

Experimental methods

Four oriented rock samples were collected from fresh pit walls at each of 38 sites that were chosen to provide regional coverage of the total of 40 sites (Fig. 1). Most are from the Main Trend because the North Trend pits are largely flooded. Sphalerite- and galena-rich ore sites were sampled from both tabular and chimney-like "prismatic" ore deposits. Several types of dolostone host rocks were also sampled. In addition, 10 oriented samples were collected from collapse breccia clasts at each of the two remaining sites (i.e., 29 and 36). A total of 509 specimens were cored from the 172 block samples. Each specimen was described petrologically and its specific gravity determined to help estimate the ore grade. Large unoriented crystals of several minerals were also collected. Details of the experimental equipment and procedures are well described in previous papers, and are not repeated here (Pan et al. 1990; Sangster and Symons 1991; Symons and Sangster 1991, 1992, in press).

After measuring the natural remanent magnetizations (NRM) of the specimens, the major problem in this study became apparent. The NRM intensities were uniformly weak, with a median of 3.6×10^{-5} A/m. Realizing that the stable characteristic remanent magnetization (ChRM) of geologic interest is typically found in the final 5–30% of the NRM on demagnetization of the specimens (Symons and Sangster, in press) and that the sensitivity limit of the automated cryogenic magnetometer used is about 4×10^{-6} A/m, it was decided to cull the collection to those sites with several "more strongly" magnetized specimens. This resulted in a reduced collection of 230 specimens with a median intensity of 6.0×10^{-5} A/m, but still having more than one-third of the specimens with NRM intensities of $< 4 \times 10^{-5}$ A/m. Eight sites were entirely excluded from measurement, including seven dolostone and one limestone host rock sites but no ore sites, suggesting that the magnetization is carried by the mineralization.

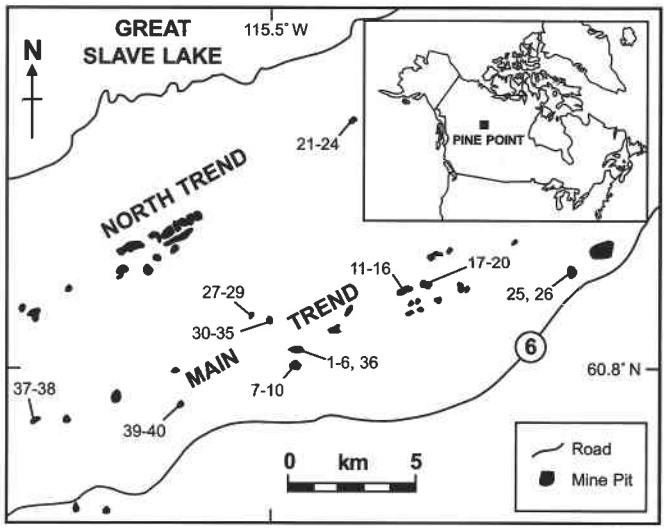


FIG. 1. Site locations at Pine Point.

The 36 specimens with the strongest NRM intensities, representing 32 sites, were demagnetized by alternating field (AF) in 11 steps to a maximum field to 110 mT. A further 12 representative specimens were thermally demagnetized in six steps to a maximum of 540°C. Ten of the 12 showed dramatic increases in remanence intensity between 400 and 500°C, signifying the oxidation of the sulphides and the generation of unwanted new magnetic minerals, and so no further thermal demagnetization was done. From past experience with MVT ores and host rocks, it was known that the ChRM is not isolated by AF demagnetization at less than 20 mT. Therefore all remaining culled specimens were AF step demagnetized at 20, 30, 40, and 50 mT.

To better characterize the magnetic minerals, 10 representative specimens were subjected to saturation isothermal remanent magnetization (SIRM) testing by magnetizing them in 13 steps to 900 mT and then AF demagnetizing them in 5 steps to 50 mT. Also, mineral cores of sphalerite (2), galena (2), calcite (1) and plaster of Paris (1) and dispersed sphalerite (1) and galena (3) in plaster of Paris were AF step demagnetized and then put through the same SIRM cycle.

Measurement results

About half of the specimens that were AF step demagnetized in detail yielded ChRM directions with mean angular deviation (MAD, Kirschvink 1980) values of less than about 10° (Fig. 2), and these directions were well dispersed despite coming from specimens with the strongest NRM intensities.

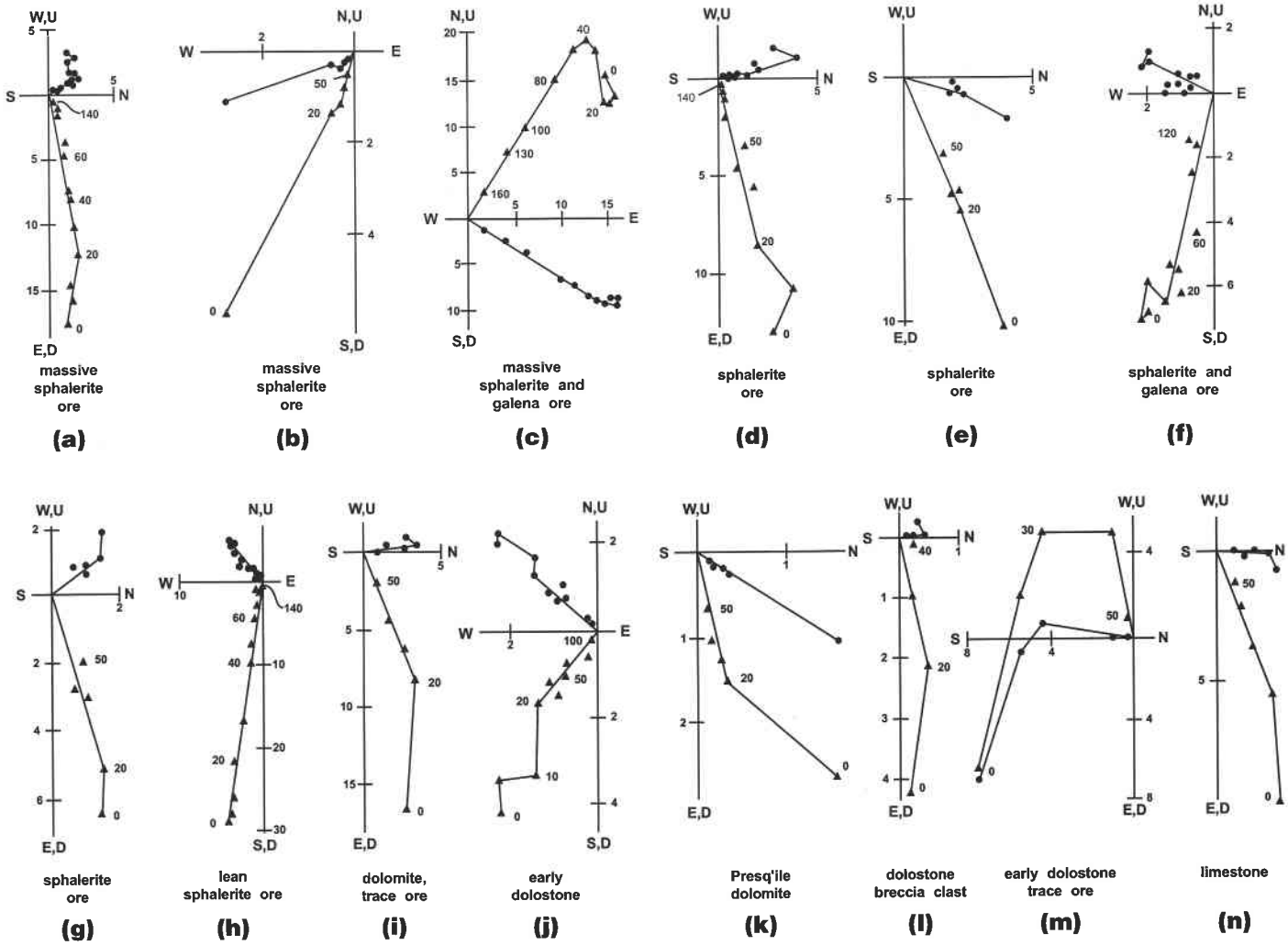


FIG. 2. Orthogonal plots showing the decay of the A ChRM vector on AF demagnetization for various ores and lithologies at Pine Point. Points in the horizontal (N, E, S, W) and vertical (U, D) planes are shown as circles and triangles, respectively. Axial intensities are in 10^{-5} A/m units. AF step field intensities are in millitesla. All panels show the normal A component except for *c* and *m*, which show the reverse A component being isolated.

The remainder of the AF step specimens give MAD values that were mostly greater than 10° with scattered directions as their remanence intensities sank below the noise level of the magnetometer.

The SIRM curves reveal some interesting features (Fig. 3). Both curves of typical grey host-rock dolostone (sites 26 and 36 in Fig. 3) are typical of pseudosingle domain (PSD) magnetite. The ore-stage brown dolostone (sites 10 and 23) and white saddle dolomite (sites 1 and 32) curves indicate either single domain (SD) magnetite (sites 23 and 32) or SD magnetite with some hematite (sites 1 and 10). The magnetization of the brown dolostone is one-half to one order of magnitude lower than for the white dolomite and grey dolostone, suggesting that it is unlikely to retain a measurable ChRM. The SIRM curves for all four ore specimens, ranging from 12 to 80% combined sphalerite and galena, indicate SD magnetite as the main magnetic mineral constituent. It is important to note that their SIRM intensities are one-half to one order of magnitude greater than for the typical grey host-rock dolostone or for the ore-associated white saddle dolomite. This implies that the remanence of the ore samples is carried predominantly by the sulphide minerals.

The mineral cores provide further insights. The two sphalerite cores have NRM intensities of 7.6 and 13.8 $\times$ 10^{-4} A/m which is about five to seven times the values of 1.7 and 1.9 $\times$ 10^{-4} A/m for the two galena cores and about 100 times greater than the value of 7.1 $\times$ 10^{-6} A/m for the calcite core. This further supports the notion that the remanence of the ore specimens is carried primarily by the sulphides, especially sphalerite. For the sphalerites, the remanence directions and intensities remain essentially constant up to 100 mT on AF demagnetization, implying SD magnetite or hematite. Note, however, that the SIRM curves for these specimens indicate that SD magnetite is predominant (Fig. 3). Because of its high conductivity, galena cannot be directly AF demagnetized or tested by SIRM. Therefore parts of the galena crystals were ground into a fine powder and allowed to acquire a detrital remanent magnetization in a plaster of Paris slurry before setting and SIRM analysis. The results from these galena and plaster cores were corrected using data from a plaster of Paris blank core. The net SIRM curves for the galena indicate that SD to PSD magnetite is present (Fig. 3). Note again that the SIRM intensities at 900 mT are higher for the sphalerite and galena than for the disseminated ores.

From the above results it is evident that the measurable remanence in the Pine Point specimens originates mostly from the ore-stage sulphides and dolomite.

Remanence directions analysis

Initial scanning of the remanence directions from all AF step demagnetized specimens, i.e., ore and host rocks, revealed considerable dispersion at the site level, so the conventional tiered statistical analysis used in paleomagnetism appeared unpromising.

Cluster analysis

The first approach to obtain a result was to use cluster analysis on the vector directions subtracted between successive demagnetizing steps following Robin and Jowett (1986). The method follows statistical theory originally developed by Kamb (1959) for hemispherical structural data and first applied to paleomagnetism by Stupavsky and Symons (1982). Robin and Jowett (1986) modified the method for use with polar paleomagnetic data. They apply computer fitting of Gaussian density surfaces to better define anomalous concentrations of vector directions and to isolate and characterize components by sequentially "erasing" the dominant component to highlight the residual components (Jowett and Pearce 1986).

When all the vectors removed (VR) are plotted for the 0–20 mT interval on AF step demagnetization for the Pine Point ores and host rocks, they form a single population that is well clustered around the present Earth's magnetic field (EMF) direction (Fig. 4*a*). At first glance this appears to be a clear signal that viscous remanent magnetization (VRM) from the present EMF is being removed from the low coercivity or soft domains. However, when the VR from 20 to 50 mT are plotted (Fig. 4*b*), it is apparent that the most prominent or A-component mode remains close to the present EMF direction. The A mode is a statistically significant cluster of directions with a mean of declination $D = 0°$, inclination $I = 75°$, peak value Pk = 14.0, and precision parameter $k = 51$ (Fisher 1953). Clusters of demonstrably less significance can be found in the elongation of the population that trends in the 135 ± 75° direction. The pole position for the A mode is located at 89°N, 114°W, (radius of cone of 95% confidence $\alpha_{95} = 18°$), similar to the Recent time-averaged EMF direction. Stripping away the 20–30 mT VR to leave the 30–50 mT VR (Fig. 4*c*) results in a significant weakening of the A peak but results in only a slight change in the pole by about 5° in the early Tertiary direction. Note, however, that the A peak remains the most significant after the application of AF demagnetization intensities that are consistent with virtually all VRM being removed in previous studies of similar MVT mineralization and their host rocks (Symons and Sangster, in press). Note, also, that the elongation in the 135° direction has also been both significantly weakened and somewhat further extended. As will be shown, the A mode is thought to be a composite population comprised of the last vestiges of VRM from the present EMF and the very weak normal A ChRM of the Pine Point ores and host rocks that are being measured as their intensities sink into the noise level of the magnetometer. Further, it will be shown that the elongation in the 135° direction probably represents VR directions from sites that have a reversed A ChRM that is too weak to be fully resolved.

An important point to make is that cluster analysis does not identify a statistically significant cluster of VR directions that corresponds to any part of the Devonian to Early Cretaceous apparent polar wander path (APWP) (Fig. 4*c*). This indicates that no primary ChRM of sedimentary origin is preserved, nor is there any secondary ChRM prior to Late Cretaceous.

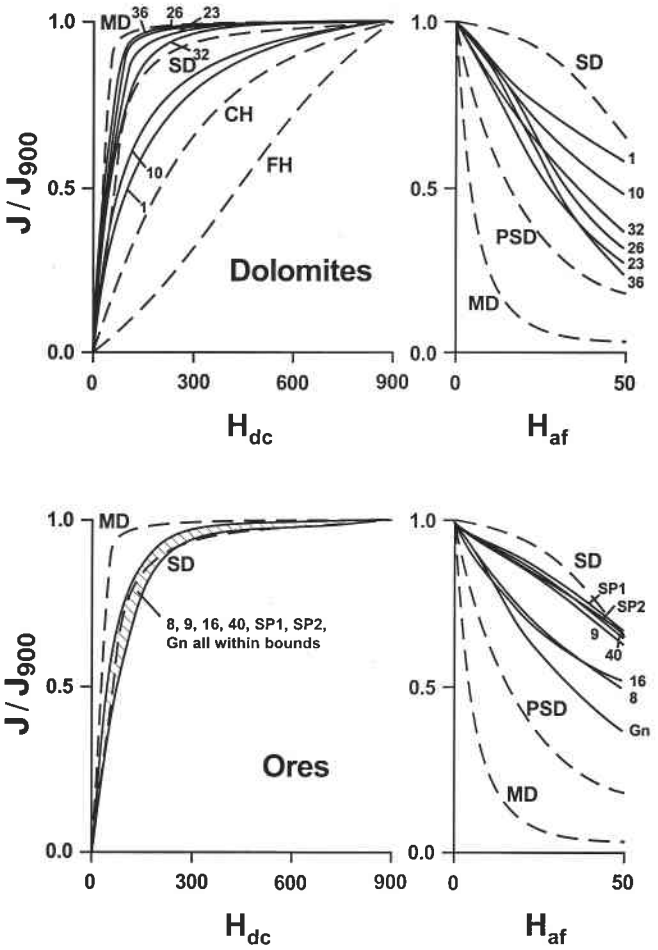


FIG. 3. Saturation isothermal remanent magnetization (SIRM) curves showing SIRM acquisition in direct fields (H_{dc}) up to 900 mT and then SIRM decay in alternating fields (H_{af}) up to 50 mT, with the magnetization shown as a ratio of the intensity acquired at 900 mT (J/J_{900}). The specimens include grey (sites 26 and 36) and brown (sites 10 and 23) dolostone and white "saddle" Presqu'ile (sites 1 and 32) dolomite. The ore specimens include disseminated ore (site 8, about 6% sphalerite and 6% galena), high-grade ore (sites 9 and 16, both about 40% sphalerite with minor galena), massive ore (site 40, about 80% sphalerite), colloform sphalerite (SP1 and SP2, both nearly pure sphalerite). and "pure" galena (Gn, about 30% galena crystal powder in plaster of Paris and corrected using plaster blank values). The J_{900} values in 10^{-2} A/m are as follows: 1, 5.1; 8, 2.9; 9, 21.5; 10, 0.57; 16, 10.2; 23, 0.54; 26, 3.3; 32, 1.1; 36, 2.4; 40, 8.5; SP1, 10.5; SP2, 34.0; and Gn, 19.1. The type curves are for single domain (SD), pseudosingle domain (PSD) and multidomain (MD) magnetite and for coarse- and fine-grained hematite (CH and FH, respectively).

Modified tiered analysis

The second approach was to relax the conventional tiered statistical approach of paleomagnetism. For each specimen the AF step remanence intensities were first checked beginning at 20 mT, and the MAD angle was calculated for the two step directions preceding the first increase in intensity using the origin as the third point. This was done on the rationale that the increase signalled the onset of acquisition of anhysteretic

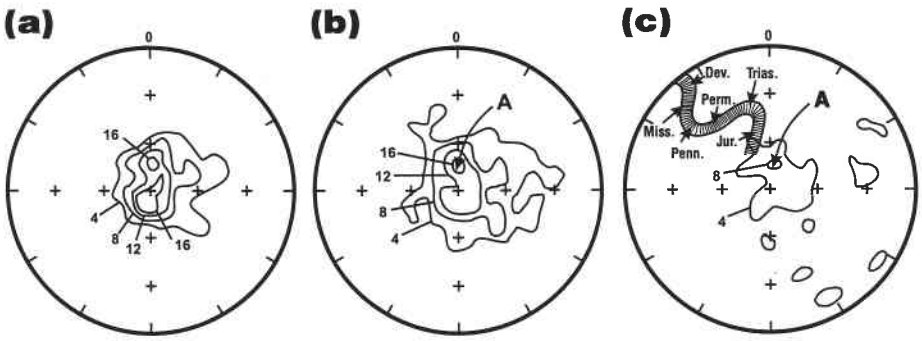


FIG. 4. Equal-area stereograms of the lower hemisphere showing density contour plots of the vectors removed (VR) for Pine Point ore and host-rock specimens between (*a*) 0 and 20 mT for 281 VR, (*b*) 20 and 50 mT for 823 VR, and (*c*) 30 and 50 mT for 523 VR. The contour intervals are expressed in increments of the circular standard deviation parameter, σ, above zero for the given N value. A is the location of the most significant mode. In *c* is shown the expected directional track for the apparent polar wander path from Middle Devonian through Jurassic time from Van der Voo (1990, 1993).

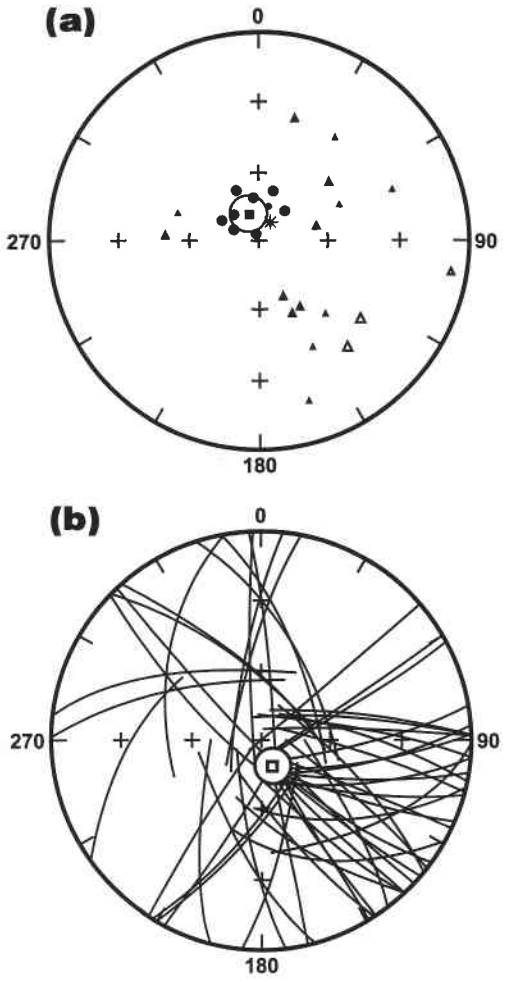


FIG. 5. Equal-area stereograms. (*a*) Accepted site mean directions with larger symbols indicating those with $\alpha_{95} < 20°$ or $k > 20$. Circles denote sites with the normal A component, and triangles denote aberrant sites with mixed VRM and reverse A magnetization. The mean A direction is shown by a square with its cone of 95% confidence. Closed (open) symbols are plotted in the lower (upper) hemisphere. The asterisk denotes the present Earth's magnetic field direction at Pine Point. (*b*) Normal A ChRM in its reversed position with specimen remagnetization circles tracking from the perimeter towards that reversed direction, suggesting that many of the deviant specimens and sites have a reversed A magnetization that has not been isolated because of their weak NRM intensities.

remanent magnetization or of instrumental noise. Three sites giving specimen directions with clearly reversed polarities (sites 9, 16, and 40) were excepted from this acceptance criterion. Specimen remanence directions with MAD angles of $< 15°$ were accepted, and they were combined to give the site mean ChRM directions listed in Table 1. The average AF demagnetizing intensity for the 103 accepted specimens is 37.8 mT, which is sufficiently intense to isolate a geologically significant ChRM.

The resulting site mean directions are not particularly well clustered internally, with all having α_{95} values of greater than 8° and only two sites having k values over 41. Reliable site mean ChRM directions were identified by accepting those with $\alpha_{95} < 20°$ or $k > 20$,thereby identifying 17 acceptable sites (Fig. 5*a*). Of these, eight site means (closed circles in Fig. 5*a*) form a well-defined cluster, defining a ChRM that is close to the present EMF direction but giving an average that is significantly different from it. This ChRM is termed the A component for the Pine Point ores and host rocks. Its mean direction is $D = 339.4°$, $I = 77.9°$ ($N = 8$, $\alpha_{95} = 7.1°$, $k = 62$). Many individual specimens retain the A component on AF demagnetization (Fig. 2), including sphalerite and galena-rich ore, disseminated ore, dolostone, and limestone. This shows that the host rocks were remagnetized when the MVT mineralization was emplaced. The eight A sites include three well-mineralized ore sites (sites 4, 17, and 18), two low-grade ore sites (sites 12 and 34), one Presqu'ile dolomite site (site 10), one dolostone site with trace mineralization (site 15) and one dolostone breccia site (site 29). The result from the breccia site indicates that the A component was acquired after brecciation. Note that a combination of VRM from the present EMF and the A ChRM in about equal proportions gives the A mode identified by cluster analysis. An obvious concern is that the other 10 accepted site mean directions are scattered, mostly along the 135° elongation direction with their ChRM swinging around to intermediate-upward inclinations. This suggested that imperfectly isolated reversed A ChRM components were present. To test this hypothesis, the AF demagnetization data were searched for specimens giving good remagnetization circles that track from a normal down direction to a reversed up direction. A total of 51 tracks from 15 sites were identified, including only 1 track from accepted site 18 and 5 tracks from rejected specimens from accepted site 29. What is obvious is that these tracks project to form a concentration of intersections in the reversed A ChRM direction

TABLE 1. Paleomagnetic results by site

Site No.[a]	Lithology[b]	Specimen vectors[a]			Mean remanence direction					
		N	N_e	N_u	D (°)	I (°)	k	α_{95} (°)	Polarity[d]	Evaluation[e]
1	Ohg	8	6	3	275	54	27	24	n	a
2	Ohs	2	0	0						
3A	Ohs	8	8	4	161	67	31	17	n	a
3B	Ohs			3	17	40	82	14	n	a
4	Omsg	9	7	3	40	76	33	22	n	aA
5	Omsg	8	8	5						
9	Ohs	8	7	4	100	−11	20	21	r	
10	Dp	8	8	5	323	87	19	18	n	aA
11A	Ols	9	8	3	66	57	27	31	n	
11B	Ols			3	290	57	7	49	n	
12	Ols	9	9	8	351	73	40	9	n	aA
13	Ohs	9	9	4	141	53	17	23	n	
14	Ols	9	9	4	14	77	13	26	n	
15	Dtsg	9	7	7	315	76	16	15	n	aA
16	Oms	7	5	3	141	−36	40	20	r	a
17	Oms	9	9	8	335	69	16	14	n	aA
18	Oms	10	7	6	14	70	17	17	n	aA
19	Ols	9	9	0						
20	Oms	9	7	7	50	55	11	19	n	a
23	D'e	5	4	0						
24	Osb	8	6	3	154	60	41	19	n	a
26	L	13	13	11	76	68	26	9	n	a
27	D'ets	9	6	0						
28	D'ets	8	4	0						
29	D'xts	17	11	6	294	79	50	10	n	aA
31	Dp	9	7	6	155	60	25	12	n	a
32	L	4	3	3	37	40	12	37	n	
34	Ols	8	7	4	297	74	134	8	m	aA
35A	Ohs	10	9	3	70	35	16	32	n	
35B	Ohs			3	165	24	7	49	n	
36	Dlx	6	0	0						
38	D'wx	4	4	0						
39	Omsg	8	6	4[f]						
40	Ohsg	9	9	6	129	−41	25	14	r	a
A mean (N = 8)					339.4	77.9	62	7.1	n	

NOTES: See Fig. 1 for site locations.

[a]A and B designate two populations of directions that were both found in two or more sample blocks.

[b]D, dolomite; D', dolostone; L, limestone; O, ore; h, high; m, moderate; l, low; t, trace; b, bitumen; c, calcite; e, early; g, galena; p, Presqu'ile; s, sphalerite; w, Watt Mountain Formation; x, breccia clasts. The lithologies for sites that provided no usable data are as follows: 6, L; 7, Dp; 21, Dp; 22, Dp, 25, D'e; 30, D'e; 33, Dp; 37, Dpc.

[c]N, N_e, and N_u, number measured, number giving acceptable end points, and number of end points used to calculate the mean, respectively.

[d]n, normal; r, reverse; m, mixed.

[e]a, sites meeting the $\alpha_{95} < 20°$ or $k > 20$ acceptance criteria; A, inclusion of the site mean in the unit mean (A mean) at the bottom of the table.

[f]Data rejected because all four specimens came from one block sample.

(Fig. 5*b*), suggesting that several reversed A sites would have been identified if sufficient NRM intensity had been present to allow more intense AF demagnetization. A very few individual specimens appear to define a reversed A component (Figs. 2*c* and 2*m*). The fact that many tracks sweep through the "southeast" quadrant provides the explanation for the elongation along the 135° trend defined by the cluster analysis.

As before, it is important to note that none of the accepted site mean directions form a group with a Devonian through mid-Cretaceous direction (Figs. 4*c* and 5*a*). This indicates that any geologically significant ChRM direction is constrained to be a Late Cretaceous or younger secondary magnetization.

Finally, it is clear that the collapse breccia clasts at site 29 also retain the A ChRM direction. This is a negative breccia or "conglomerate" test which shows that this A remanence is secondary and acquired after formation of the breccia.

Pole position

The pole position calculated for the A ChRM is 79.2°N, 196.9°E (α_{95} = 12.9°). This pole lies between the Early Cretaceous and Oligocene reference poles on the APWP for the North American craton (Fig. 6) and is statistically distinct from them at the 95% confidence level. Although it is not statistically distinct from either the Late Cretaceous or Eocene reference poles at the 95% level, it is centred on the mid-Late Cretaceous through Paleocene reference pole (84−58 Ma) which corresponds in time to the Laramide orogeny (97−

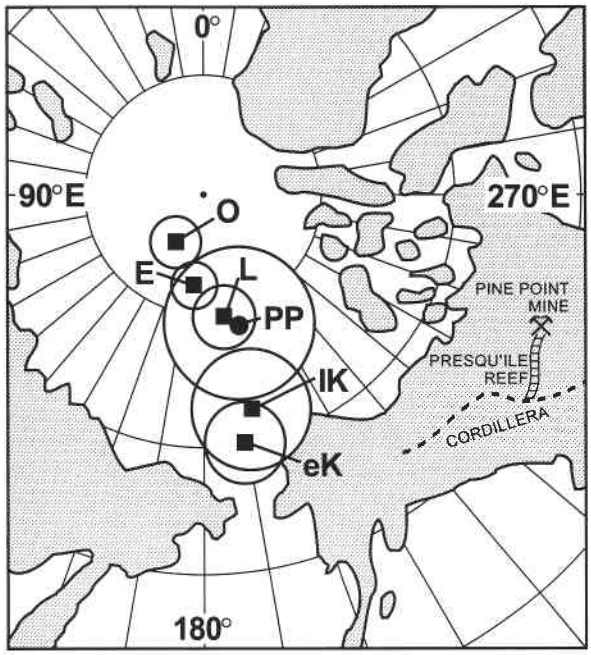


FIG. 6. The Pine Point ore and host-rock A magnetization component paleopole position (●, PP) with respect to reference poles (■) for the North American craton from rock units of Early Cretaceous (eK, 71°N, 190°E, α_{95} = 5.4, Van der Voo 1993), Late Cretaceous (lK, 73°N, 193°E, α_{95} = 9.7°, Van der Voo 1993), mid-Late Cretaceous to Paleocene (L, Laramide orogeny; 80°N, 190°E, α_{95} = 4.2°, Wynne et al., 1992), Eocene (E, 83°N, 176°E, α_{95} = 2.9°, Irving and Brandon 1990) and Oligocene (O, 86°N, 154°E, α_{95} = 3.0°, Irving and Irving 1982) age. The poles are surrounded by their standard error circles (P = 0.37), which are about half of the radius of the 95% error circles (P = 0.05). To a close approximation paleopoles are significantly different at the 95% confidence level if their standard error circles do not overlap.

58 Ma). During this time the Omineca and Foreland belts of the Cordillera underwent as much as 200 km of crustal shortening to uplift the Front Ranges across the Western Canada sedimentary basin from Pine Point (Gabrielse and Yorath 1992). Had the A ChRM of the Pine Point ores and host rocks been more intense, its direction would have been much more precisely defined. Recalling the abundant evidence of reversed A ChRM from the remagnetization circles, it is unlikely that the ChRM is older than mid-Late Cretaceous, which is at the end of the Cretaceous normal superchron of about 35 Ma duration. Recalling also the abundant evidence from the rock magnetic tests on monomineralic specimens that galena and sphalerite are the main ChRM carriers, it is evident that the measured pole position records the age of host-rock remagnetization and MVT mineralization.

The results of this paleomagnetic study support only the 60 Ma age for the Pine Point ore obtained by Cumming et al. (1990) using a three-stage lead-evolution model. It also supports the geological arguments of Qing and Mountjoy (1992) from strontium- and oxygen-isotope values and of Arne (1991) from apatite fission-track studies that the only time that the host rocks could have been sufficiently heated since deposition to give the 85–100°C temperatures recorded in the fluid inclusions by Roedder (1968) and Kyle (1981) was during the Late Cretaceous or early Tertiary, coeval with the Laramide orogeny. The Laramide age found in this study conflicts with the Permian mineralization age reported in an abstract by Beales and Jackson (1982). Only two sites (sites 16 and 40) for Pine Point give mean directions that are typical of a Permian remanence, but both of them also give several remagnetization circles (4 and 8, respectively) that are clearly heading to the steeper inclinations expected for a reversed A ChRM.

The finding of a Laramide age for the Pine Point sphalerite–galena mineralization has consequences for the various genetic models that have been proposed for these deposits. Jackson and Beales (1967) suggested that MVT deposits were formed by steady-state compaction-driven flow of the ore fluids out of adjacent clastic sediments during basin infilling by overlying sediments, with subsequent MVT ore deposition in the adjacent carbonate platform. Thus they argued for a late Paleozoic age for the Pine Point ores. Cathles and Smith (1983), in order to reach the required ore temperatures, argued for episodic compaction-driven flow of the ore fluids. From their calculations they suggested that ore formation would occur within a few tens of millions of years of host-rock deposition. Thus their model predicts a late Paleozoic age for the Pine Point deposits. Clearly our result is opposed to both of these compaction-driven fluid-flow models. In contrast our result strongly supports gravity-driven fluid-flow models.

The coincidence in timing of Pine Point mineralization with that of tectonic uplift of the eastern Cordillera during the Laramide orogeny lends support for the model for MVT deposits proposed by Garven and Freeze (1984*a*, 1984*b*), Bethke (1985), and Oliver (1986). Gravity-driven fluid flow, produced by uplift in the Rocky Mountains west of Pine Point, would by this model force basinal brines eastward and eventually into the carbonate platform, resulting in sulphide deposition at Pine Point (Fig. 6). Indeed, Garven (1985) specifically predicted a Laramide age for the Pine Point deposits with the fluids being driven eastward out of the Western Canada sedimentary basin by the Laramide orogeny. Rowan and Leach (1989) extended the gravity-induced fluid-flow model to suggest that ore deposition would occur in the foreland bulge produced by uplift of the adjacent basin edge. For Pine Point it is unclear whether the deposits are in the foreland bulge. Rather, they are spatially related to the Presqu'ile barrier reef at the basin's margin (Fig. 6). However, Kesler et al. (1972) showed from asymmetrical crystal growth that the Pine Point mineralizing fluids flowed mainly from west to east. Qing and Mountjoy (1992) have recently summarized considerable geochemical evidence to also show that hot radiogenic basinal fluids migrated eastwards up dip along the Presqu'ile barrier, thereby causing extensive dolomitization, secondary migration of hydrocarbons and MVT mineralization in the Pine Point area.

Conclusions

Paleomagnetic analysis of the Pine Point Zn–Pb deposits and their dolostone host rocks shows that their ChRM is about an order of magnitude weaker than those for other MVT districts studied to date. The remanence and SIRM measurements show that the ChRM is carried dominantly by sphalerite and galena, and that it has the attributes of typical single domain to pseudosingle domain magnetite. A breccia test shows that the ChRM was acquired after formation of the collapse breccias in the host rocks. Eight of the 17 sites for which a reasonably well-clustered site mean direction could be determined form a coherent cluster or A magnetization component direction. Its unit mean direction gives a pole position of 79°N,

197°E ($N = 8$, $\alpha_{95} = 13°$) that is coincident at the 95% confidence level with Late Cretaceous to Eocene positions for the North American APWP, but that agrees best with mid-Late Cretaceous to Paleocene poles (84–58 Ma) coincident with the Laramide orogeny (97–58 Ma). This age for the Pine Point Zn–Pb mineralization event agrees with indirect age dating evidence based on thermal and geological considerations. Until these ores are directly dated by some other method, this pole should not be used for APWP construction. The age provides support for gravity-driven fluid-flow models and argues against compaction-driven fluid-flow models for the genesis of MVT deposits.

Acknowledgments

The authors thank Cominco Ltd. for permission to work at Pine Point; A. Keen, Manager, Polaris Mines, for arranging for sampling after closure of the mine site; O.E. Owens, Vice-President Exploration, for permission to publish the results; T.B. Symons for preparing the samples; D.J. Alldrick, P.J. Wynne, and an anonymous referee for useful suggestions to improve the manuscript; and the Department of Energy, Mines and Resources Canada and the Natural Sciences and Engineering Research Council of Canada who funded this research.

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